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CONNECTOR DEVICE AND ELECTRIC CONNECTOR

Abstract

A connector device includes an electric connector and a mating connector fitted to the electric connector. The electric connector includes a pair of first contacts, a first shell partially covering a periphery of the pair of first contacts around an axis along a fitting direction between the electric connector and the mating connector, and a first housing. The mating connector includes a pair of second contacts, a second shell partially covering a periphery of the pair of second contacts around an axis, and a second housing. When viewed from the fitting direction in the fitted state where the electric connector and the mating connector are fitted to each other, the first shell and the second shell surround a contact touching portion where the pair of first contacts and the pair of second contacts come into contact with each other around an axis.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application of PCT Application No. PCT/JP2023/040607, filed on Nov. 10, 2023, which claims the benefit of priority from Japanese Patent Application No. 2022-202000, filed on Dec. 19, 2022. The entire contents of the above listed PCT and priority applications are incorporated herein by reference.

BACKGROUND
[0002] The present disclosure relates to a connector device and an electric connector.
[0003] Japanese Patent No. 6965685 discloses a receptacle connector that is mounted on a board and is fitted to a plug connector connected to a tip of a cable. The receptacle connector of Japanese Patent No. 6965685 includes a shell connected to a shell of a plug connector while covering a pair of contact members.
[0004] There has been a demand for simplifying a shell of an electric connector mounted on a board.
SUMMARY
[0005] A connector device according to one aspect of the present disclosure includes an electric connector mounted on a first board and a mating connector mounted on a second board and fitted to the electric connector. The electric connector includes a pair of conductive first contacts electrically connected to a signal conductor of the first board, a conductive first shell electrically connected to a ground conductor of the first board and partially covering a periphery of the pair of first contacts around an axis along a fitting direction between the electric connector and the mating connector, and a first housing configured to hold the pair of first contacts and the first shell in an insulated state. The mating connector includes a pair of conductive second contacts electrically connected to a signal conductor of the second board, a conductive second shell electrically connected to a ground conductor of the second board and partially covering a periphery of the pair of second contacts around the axis, and a second housing configured to hold the pair of second contacts and the second shell in an insulated state. The pair of first contacts comes into contact with the pair of second contacts and the first shell comes into contact with the second shell in a fitted state where the electric connector and the mating connector are fitted to each other. The first shell and the second shell surround a contact touching portion where the pair of first contacts and the pair of second contacts come into contact with each other around the axis when viewed from the fitting direction in the fitted state.
[0006] In the electric connector, the first shell partially covers the periphery of the pair of first contacts around the axis. This enables simplifying the first shell as compared with a case where the first shell covers the entire periphery of the pair of first contacts around the axis. In the mating connector, the second shell partially covers the periphery of the pair of second contacts around the axis. This enables simplifying the second shell as compared with a case where the second shell covers the entire periphery of the pair of second contacts around the axis. When viewed from the fitting direction in the fitted state, the first shell and the second shell surround the contact touching portion around the axis. This improves the shielding performance in the fitted state.
[0007] The first shell may include a pair of first side surfaces facing each other and a first coupling surface coupling the pair of first side surfaces. The second shell may include a pair of second side surfaces facing each other and a second coupling surface coupling the pair of second side surfaces.

The pair of first side surfaces, the pair of second side surfaces, the first coupling surface, and the second coupling surface may surround the contact touching portion around the axis when viewed from the fitting direction in the fitted state. The first shell has a so-called U-shape or H-shape with the pair of first side surfaces and the first coupling surface. The second shell has a so-called U-shape or H-shape with the pair of second side surfaces and the second coupling surface. Such shapes of the first shell and the second shell may be readily manufactured by, for example, performing bending processing or the like on a punched metal plate. This enables simplifying the first shell and the second shell. Further, when viewed from the fitting direction in the fitted state, the pair of first side surfaces, the pair of second side surfaces, the first coupling surface, and the second coupling surface surround the periphery of the contact touching portion around the axis. This improves the shielding performance in the fitted state.

[0008] The pair of second side surfaces may include a pair of protrusions protruding away from the second board. The pair of protrusions may come into contact with the pair of first side surfaces in the fitted state. A shell touching portion where the pair of protrusions and the pair of first side surfaces come into contact with each other may be closer to the first board than the contact touching portion. In the fitted state, the pair of protrusions is in contact with the pair of first side surfaces at a position closer to the first board than the contact touching portion. The distance between the second board and the first board is reduced and the length of the pair of protrusions is increased in the direction from the second board toward the first board, which increases the elastic force of the pair of protrusions.

[0009] When viewed from the fitting direction in the fitted state, the pair of first contacts may be electrically connected to the signal conductor of the first board outside a region surrounded by the first shell and the second shell. In the region surrounded by the first shell and the second shell, the pair of first contacts comes into contact with the pair of second contacts. The pair of first contacts is electrically connected to the signal conductor of the first board outside the region surrounded by the first shell and the second shell. That is, the pair of first contacts is formed both inside and outside the region surrounded by the first shell and the second shell. In such a pair of first contacts, the length from a contact point with the pair of second contacts to a contact point with the signal conductor of the first board increases, so that the elastic force of the pair of first contacts may be increased.

[0010] The electric connector may include a plurality of first units each of which includes the pair of first contacts and the first shell. The plurality of first units may be disposed along a direction orthogonal to the fitting direction and a direction orthogonal to an arrangement direction of the pair of first contacts, and may be disposed along the arrangement direction of the pair of first contacts. The mating connector may include a plurality of second units each of which includes the pair of second contacts and the second shell. The plurality of second units may be disposed along a direction orthogonal to the fitting direction and a direction orthogonal to an arrangement direction of the pair of second contacts, and may be disposed along the arrangement direction of the pair of second contacts. In the fitted state, the shielding performance may be achieved for each of the plurality of first units and the plurality of second units. As a result, the interval between the first units and the interval between the second units may be reduced, which enables the electric connector and the mating connector to achieve higher density.

[0011] The pair of first contacts may include a first connection leg portion extending along the first board and electrically connected to the signal conductor of the first board. The pair of second contacts may include a second connection leg portion extending along the second board and electrically connected to the signal conductor of the second board. In the fitted state, the first connection leg portion and the second connection leg portion may extend in directions away from each other. The first connection leg portion and the second connection leg portion extend in directions away from each other. Since the first connection leg portion and the second connection leg portion are spaced apart from each other, it may prevent the influence that the first connection leg portion and the second connection leg portion have on each other.

[0012] The pair of first side surfaces may face each other with the first connection leg portion interposed therebetween in an arrangement direction of the pair of first contacts. The pair of second side surfaces may face each other with the second connection leg portion interposed therebetween in an arrangement direction of the pair of second contacts. In the fitted state, the pair of first side surfaces may face each other with the contact touching portion interposed therebetween in the arrangement direction of the pair of first contacts, and the pair of second side surfaces may face each other with the contact touching portion interposed therebetween in the arrangement direction of the pair of second contacts. This improves the shielding performance in the arrangement direction of the pair of first contacts and the pair of second contacts.

[0013] In the fitted state, the first coupling surface may not face the second connection leg portion. The second coupling surface may not face the first connection leg portion. In the fitted state, the first coupling surface opens the second connection leg portion, and the second coupling surface opens the first connection leg portion. This improves the visibility of the first connection leg portion and the second connection leg portion.

[0014] When viewed from the fitting direction in the fitted state, the pair of first contacts may pass between the second coupling surface and the first board, extend outside a region surrounded by the first shell and the second shell, and be electrically connected to the signal conductor of the first board. The pair of first contacts is formed to extend between the second coupling surface and the first board. That is, the pair of first contacts is formed both inside and outside the region surrounded by the first shell and the second shell. In such a pair of first contacts, the length from a contact point with the pair of second contacts to a contact point with the signal conductor of the first board increases, so that the elastic force of the pair of first contacts may be increased.

[0015] In the fitted state, the first coupling surface and the second coupling surface do not have to face each other when viewed from a direction orthogonal to the fitting direction and a direction orthogonal to an arrangement direction of the pair of first contacts. In this case, the visibility of the pair of first contacts and the pair of second contacts is increased.

[0016] The first housing may be formed with a fitting hole penetrating the first housing so as to connect a main surface facing the mating connector when fitted to the mating connector and a back surface facing the first board. The second housing may be formed with a fitting protrusion protruding away from the second board with a main surface facing the electric connector when fitted to the electric connector as a base end. The fitting hole and the fitting protrusion may be fitted to each other in the fitted state. In this case, the fitting protrusion and the fitting hole function as guides at the time of fitting the electric connector and the mating connector. As a result, the reliability of the contact between the first contact and the second contact may be increased.

[0017] An electric connector according to one aspect of the present disclosure is mounted on a first board and is fitted to a mating connector mounted on a second board. The electric connector includes a pair of conductive first contacts electrically connected to a signal conductor of the first board, a conductive first shell electrically connected to a ground conductor of the first board and partially covering a periphery of the pair of first contacts around an axis along a fitting direction between the electric connector and the mating connector, and a housing configured to hold the pair of first contacts and the first shell in an insulated state. The pair of first contacts comes into contact with a pair of conductive second contacts of the mating connector and the first shell comes into contact with a conductive second shell of the mating connector in a fitted state with the mating connector. The first shell and the second shell surround a contact touching portion where the pair of first contacts and the pair of second contacts come into contact with each other around the axis when viewed from the fitting direction in the fitted state.

[0018] In the electric connector, the first shell partially covers the periphery of the pair of first contacts around the axis. This enables simplifying the first shell as compared with a case where the first shell covers the entire periphery of the pair of first contacts around the axis. When viewed

from the fitting direction in the fitted state with the mating connector, the first shell and the second shell surround the contact touching portion around the axis. This improves the shielding performance in the fitted state.
[0019] According to the present disclosure, a connector device and an electric connector capable of simplifying a shell may be provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0020] FIG. **1** is a perspective view illustrating an example of the appearance of a connector device.
[0021] FIG. **2** is a plan view of an electric connector.
[0022] FIG. **3**A is a perspective view illustrating an example of the appearance of a pair of first contacts. FIG. **3**B is a plan view illustrating an example of the appearance of the pair of first contacts. FIG. **3**C is a side view illustrating an example of the appearance of the pair of first contacts. FIG. **3**D is a bottom view illustrating an example of the appearance of the pair of first contacts.
[0023] FIG. **4**A is a perspective view illustrating an example of the appearance of a first shell. FIG. **4**B is a plan view illustrating an example of the appearance of the first shell. FIG. **4**C is a side view illustrating an example of the appearance of the first shell. FIG. **4**D is a bottom view illustrating an example of the appearance of the first shell.
[0024] FIG. **5**A is a perspective view illustrating an example of the appearance of a first unit. FIG. **5**B is a plan view illustrating an example of the appearance of the first unit. FIG. **5**C is a side view illustrating an example of the appearance of the first unit. FIG. **5**D is a bottom view illustrating an example of the appearance of the first unit.
[0025] FIG. **6** is an enlarged plan view illustrating an example of a part of the appearance of an electric connector in an enlarged manner.
[0026] FIG. **7** is an enlarged bottom view illustrating an example of a part of the appearance of the electric connector in an enlarged manner.
[0027] FIG. **8** is a side view illustrating an example of coupling by a first side shell.
[0028] FIG. **9** is a plan view of a mating connector.
[0029] FIG. **10**A is a perspective view illustrating an example of the appearance of a pair of second contacts. FIG. **10**B is a plan view illustrating an example of the appearance of the pair of second contacts. FIG. **10**C is a side view illustrating an example of the appearance of the pair of second contacts. FIG. **10**D is a bottom view illustrating an example of the appearance of the pair of second contacts.
[0030] FIG. **11**A is a perspective view illustrating an example of the appearance of a second shell. FIG. **11**B is a plan view illustrating an example of the appearance of the second shell. FIG. **11**C is a side view illustrating an example of the appearance of the second shell. FIG. **11**D is a bottom view illustrating an example of the appearance of the second shell.
[0031] FIG. **12**A is a perspective view illustrating an example of the appearance of a second unit. FIG. **12**B is a plan view illustrating an example of the appearance of the second unit. FIG. **12**C is a side view illustrating an example of the appearance of the second unit. FIG. **12**D is a bottom view illustrating an example of the appearance of the second unit.
[0032] FIG. **13** is an enlarged plan view illustrating an example of a part of the appearance of a mating connector in an enlarged manner.
[0033] FIG. **14** is an enlarged bottom view illustrating an example of a part of the appearance of the mating connector in an enlarged manner.
[0034] FIG. **15** is a side view illustrating an example of coupling by a second side shell.
[0035] FIG. **16** is a perspective view illustrating an example of the appearance of the first unit and

the second unit in a fitted state.
[0036] FIG. **17** is a plan view illustrating an example of the appearance of the first unit and the second unit in a fitted state.
[0037] FIG. **18** is a cross-sectional view illustrating a cross section taken along the line xviii-xviii of FIG. **17**.
[0038] FIG. **19** is a cross-sectional view illustrating a cross section taken along the line xix-xix of FIG. **17**.
[0039] FIG. **20** is a perspective view illustrating an example of the appearance of a connector device according to a modification.
[0040] FIG. **21**A is a perspective view illustrating an example of the appearance of a first unit according to a modification. FIG. **21**B is a plan view illustrating an example of the appearance of the first unit according to the modification. FIG. **21**C is a side view illustrating an example of the appearance of the first unit according to the modification. FIG. **21**D is a bottom view illustrating an example of the appearance of the first unit according to the modification.

DETAILED DESCRIPTION

[0041] In the following description, with reference to the drawings, the same reference numbers are assigned to the same components or to similar components having the same function, and overlapping description is omitted.

Connector Device

[0042] FIG. **1** is a perspective view illustrating an example of the appearance of a connector device **1** according to the present disclosure. The connector device **1** is, for example, a device used for connecting printed circuit boards in an electronic device. The connector device **1** includes an electric connector **2** mounted on a first board **200** (see FIG. **2**) and a mating connector **3** mounted on a second board **300** (see FIG. **9**) and fitted to the electric connector **2**. In the connector device **1**, the electric connector **2** and the mating connector **3** are fitted to each other along the Z-axis direction. The electric connector **2** and the mating connector **3** are fitted to each other, so that the first board **200** and the second board **300** are electrically connected. Hereinafter, the Z-axis direction is referred to as a fitting direction. When viewed from the fitting direction, each of the electric connector **2** and the mating connector **3** takes on a form approximating a rectangular shape. Hereinafter, when viewed from the fitting direction, the short side direction of the rectangular electric connector **2** and the rectangular mating connector **3** is referred to as an X-axis direction, and the long side direction thereof is referred to as a Y-axis direction. The X-axis direction, the Y-axis direction, and the Z-axis direction are orthogonal to each other.

Electric Connector

[0043] FIG. **2** is a plan view of the electric connector **2**. The electric connector **2** is, for example, a receptacle connector, and is mounted on the first board **200**. The first board **200** is, for example, a printed circuit board. The first board **200** is formed with a plurality of signal conductors (conductors forming a part of a signal circuit) (not illustrated) and a plurality of ground conductors (conductors forming a part of a ground circuit) (not illustrated). The electric connector **2** includes a pair of first contacts **21**, a first shell **22**, a first housing **23**, and a pair of first side shells **24**. The electric connector **2** includes a plurality of first units **20** each of which includes the pair of first contacts **21** and the first shell **22**.
[0044] The pair of first contacts **21** is electrically connected to the signal conductors of the first board **200**. The pair of first contacts **21** transmits a plurality of types of differential signals. Examples of the plurality of types of differential signals include a high-speed signal and a low-speed signal. The pair of first contacts **21** is disposed side by side while being spaced apart from each other along the X-axis direction. Hereinafter, the X-axis direction is also referred to as an arrangement direction.
[0045] The first shell **22** is formed of a conductive metal plate, for example, and is electrically connected to the ground conductors of the first board **200**. The first shell **22** partially covers the

periphery of the pair of first contacts **21** around an axis L along the fitting direction. For example, the first shell **22** covers the periphery of the pair of first contacts **21** from two directions in the X-axis direction and one direction in the Y-axis direction. The first shell **22** is not tubular.

[0046] The first housing **23** is formed of an insulating resin, for example, and holds therein the pair of first contacts **21** and the first shell **22** in an insulated state. The first housing **23** holds therein the plurality of first units **20**. The first housing **23** includes a plurality of blocks **230** each of which holds therein the plurality of first units **20** along the X-axis direction and which is arranged side by side along the Y-axis direction.

[0047] The plurality of first units **20** is disposed along a direction orthogonal to the fitting direction and a direction orthogonal to the arrangement direction of the pair of first contacts **21**, and is also disposed along the arrangement direction of the pair of first contacts **21**. That is, the plurality of first units **20** is disposed along the X-axis direction and the Y-axis direction. The number of first units **20** is not limited. In one example, the number of first units **20** may be 60. In this case, 10 first units **20** disposed along the X-axis direction may be arranged in 6 sets along the Y-axis direction. An interval D**1** between the plurality of first units **20** adjacent to each other in the X-axis direction may be smaller than an interval D**2** between the plurality of first units **20** adjacent to each other in the Y-axis direction.

[0048] The pair of first side shells **24** is formed of a conductive metal plate, for example, and is a coupling member that couples the plurality of blocks **230** of the first housing **23** along the Y-axis direction. The pair of first side shells **24** is disposed at both ends of the plurality of blocks **230** in the X-axis direction. The pair of first side shells **24** engages with each of the plurality of blocks **230** to couple the plurality of blocks **230**.

[0049] FIG. **3**A is a perspective view illustrating an example of the appearance of the pair of first contacts **21**. FIG. **3**B is a plan view illustrating an example of the appearance of the pair of first contacts **21**. FIG. **3**C is a side view illustrating an example of the appearance of the pair of first contacts **21**. FIG. **3**D is a bottom view illustrating an example of the appearance of the pair of first contacts **21**.

[0050] As illustrated in FIG. **3**B and FIG. **3**D, the pair of first contacts **21** extends along the Y-axis direction when viewed from the fitting direction. As illustrated in FIG. **3**C, the pair of first contacts **21** takes on a form approximating an S shape when viewed from the arrangement direction. The pair of first contacts **21** is formed by, for example, performing bending processing or the like on a punched metal plate. Each of the pair of first contacts **21** includes a first contact portion **211**, a first connection leg portion **212**, and a first coupling portion **213**.

[0051] The first contact portion **211** protrudes along the Z-axis direction so as to be spaced apart from the first board **200**. The first connection leg portion **212** extends along the first board **200** and is electrically connected to the signal conductors of the first board **200**. The first coupling portion **213** couples the first contact portion **211** and the first connection leg portion **212**. The first coupling portion **213** is formed in a crank shape by extending away from the first board **200** and then bending toward the first board **200**.

[0052] FIG. **4**A is a perspective view illustrating an example of the appearance of the first shell **22**. FIG. **4**B is a plan view illustrating an example of the appearance of the first shell **22**. FIG. **4**C is a side view illustrating an example of the appearance of the first shell **22**. FIG. **4**D is a bottom view illustrating an example of the appearance of the first shell **22**.

[0053] As illustrated in FIG. **4**B and FIG. **4**D, the first shell **22** has a so-called U-shape or H-shape when viewed from the fitting direction. The first shell **22** is formed by, for example, performing bending processing or the like on a punched metal plate. The first shell **22** includes a pair of first side surfaces **221** facing each other and a first coupling surface **222** coupling the pair of first side surfaces **221**. The first shell **22** does not have a surface facing the first coupling surface **222**.

[0054] As illustrated in FIG. **4**A, the pair of first side surfaces **221** has a flat plate shape extending in the Y-axis direction and the Z-axis direction. Each of the pair of first side surfaces **221** includes a ground connection portion **221***a* that faces the first board **200** and is formed along the Y-axis direction. The ground connection portion **221***a* is electrically connected to the ground conductors of the first board **200**.

[0055] The first coupling surface **222** has a flat plate shape extending in the X-axis direction and the Z-axis direction. The first coupling surface **222** couples the pair of first side surfaces **221** at both ends of the first coupling surface **222** in the X-axis direction. The first coupling surface **222** includes a ground connection portion **222***a* that faces the first board **200** and is formed along the X-axis direction. The ground connection portion **222***a* is electrically connected to the ground conductors of the first board **200**.

[0056] FIG. **5**A is a perspective view illustrating an example of the appearance of the first unit **20**. FIG. **5**B is a plan view illustrating an example of the appearance of the first unit **20**. FIG. **5**C is a side view illustrating an example of the appearance of the first unit **20**. FIG. **5**D is a bottom view illustrating an example of the appearance of the first unit **20**.

[0057] The pair of first side surfaces **221** faces each other with the pair of first contacts **21** interposed therebetween in the arrangement direction of the pair of first contacts **21**. The pair of first side surfaces **221** is disposed so as to cover the pair of first contacts **21** in the arrangement direction of the pair of first contacts **21**. As illustrated in FIG. **5**B, the total length of the pair of first side surfaces **221** in the Y-axis direction is longer than the total length of the pair of first contacts **21** in the Y-axis direction. As illustrated in FIG. **5**D, the total length of the ground connection portion **221***a* in the Y-axis direction is longer than the total length of the pair of first contacts **21** in the Y-axis direction.

[0058] The first coupling surface **222** faces the first contact portions **211** of the pair of first contacts **21** in the Y-axis direction. The first coupling surface **222** is disposed so as to cover the first contact portions **211** of the pair of first contacts **21** in the Y-axis direction. As illustrated in FIG. **5**D, the total length of the ground connection portion **222***a* in the X-axis direction is longer than the distance between the pair of first contacts **21** disposed in the X-axis direction.

[0059] The first housing **23** will be described with reference to FIGS. **6** and **7**. FIG. **6** is an enlarged plan view illustrating an example of a part of the appearance of the electric connector **2** in an enlarged manner. FIG. **7** is an enlarged bottom view illustrating an example of a part of the appearance of the electric connector **2** in an enlarged manner. When viewed from the fitting direction, the first housing **23** takes on a form approximating a rectangular shape. Each of the blocks **230** of the first housing **23** includes a main surface **230***a* and a back surface **230***b* that extend in a planar shape along the first board **200**.

[0060] The main surface **230***a* is a surface facing the mating connector **3** when fitted to the mating connector **3**. The back surface **230***b* is a surface facing the first board **200**. Each of the blocks **230** is formed with, for each of the plurality of first units **20**, a pair of inner grooves **231**, a window **232**, a pair of outer grooves **233**, a coupling groove **234**, a fitting hole **235**, and an engagement groove **236** that penetrate so as to connect between the main surface **230***a* and the back surface **230***b*.

[0061] The pair of inner grooves **231** is so formed that the pair of first contacts **21** is disposed therein. The window **232** is a hole that exposes the first connection leg portion **212** of the first contact **21** when viewed from the fitting direction. The pair of outer grooves **233** is so formed that the pair of inner grooves **231** is interposed therebetween in the arrangement direction of the pair of first contacts **21**. In the pair of outer grooves **233**, the pair of first side surfaces **221** of the first shell **22** is accommodated. The coupling groove **234** is a groove that couples the pair of outer grooves **233**. In the coupling groove **234**, the first coupling surface **222** of the first shell **22** is accommodated. The fitting hole **235** is fitted to a fitting protrusion **334**, described later, when the electric connector **2** and the mating connector **3** are fitted to each other. In a case where the plurality of blocks **230** is arranged in the Y-axis direction to manufacture the electric connector **2**, the fitting hole **235** enables positioning between the blocks **230** by fitting a protrusion of a jig (not illustrated) on which the blocks **230** are placed into the fitting hole **235**. In the engagement groove

236, the first side shell **24** is accommodated. One or more engagement grooves **236** may be provided per block **230**. For example, a plurality of engagement grooves **236** is provided at predetermined intervals in the Y-axis direction.

[0062] FIG. **8** is a side view illustrating an example of coupling by the first side shell **24**. The first side shell **24** has a comb-like shape. The first side shell **24** is formed of, for example, a metal plate. The first side shell **24** includes a coupling plate **241** and a protrusion **242**. The coupling plate **241** has an elongated plate-like shape and extends in a planar shape in the Y-axis direction. The protrusions **242** protrude toward the engagement grooves **236** of the plurality of blocks **230** in the fitting direction. The protrusions **242** are provided so as to correspond to the number of engagement grooves **236**. For example, the plurality of protrusions **242** is provided at predetermined intervals in the Y-axis direction. The plurality of protrusions **242** is accommodated in the plurality of engagement grooves **236** by press-fitting for each axis K**1** along the fitting direction.

Mating Connector

[0063] FIG. **9** is a plan view of the mating connector **3**. FIG. **9** illustrates a state in which the top and bottom are inverted in the fitting direction. That is, FIG. **9** corresponds to a plan view seen from the electric connector **2** along the fitting direction. The mating connector **3** is, for example, a plug connector, and is mounted on the second board **300**. The second board **300** is, for example, a printed circuit board. The second board **300** is formed with a plurality of signal conductors and a plurality of ground conductors (not illustrated). The mating connector **3** includes a pair of second contacts **31**, a second shell **32**, a second housing **33**, and a pair of second side shells **34**. The mating connector **3** includes a plurality of second units **30** each of which includes the pair of second contacts **31** and the second shell **32**.

[0064] The pair of second contacts **31** is formed of a conductive metal plate, for example, and is electrically connected to the signal conductors of the second board **300**. The pair of second contacts **31** transmits a plurality of types of differential signals. Examples of the plurality of types of differential signals include a high-speed signal and a low-speed signal. The pair of second contacts **31** is disposed side by side while being spaced apart from each other along the X-axis direction.

[0065] The second shell **32** is formed of a conductive metal plate, for example, and is electrically connected to the ground conductors of the second board **300**. The second shell **32** partially covers the periphery of the pair of second contacts **31** around an axis L along the fitting direction. For example, the second shell **32** covers the periphery of the pair of second contacts **31** from two directions in the X-axis direction and one direction in the Y-axis direction. The second shell **32** is not tubular.

[0066] The second housing **33** is formed of an insulating resin, for example, and holds therein the pair of second contacts **31** and the second shell **32** in an insulated state. The second housing **33** holds therein the plurality of second units **30**. The second housing **33** includes a plurality of blocks **330** each of which holds therein the plurality of second units **30** along the X-axis direction and which is arranged side by side along the Y-axis direction. When viewed from the fitting direction, each of the blocks **330** takes on a form approximating a concave shape.

[0067] The plurality of second units **30** is disposed along a direction orthogonal to the fitting direction and a direction orthogonal to the arrangement direction of the pair of second contacts **31**, and is also disposed along the arrangement direction of the pair of second contacts **31**. That is, the plurality of second units **30** is disposed along the Y-axis direction and the X-axis direction. The number of second units **30** is not limited. In one example, the number of second units **30** may be 60. In this case, 10 second units **30** disposed along the X-axis direction may be arranged in 6 sets along the Y-axis direction. An interval D**3** between the plurality of second units **30** adjacent to each other in the X-axis direction may be smaller than an interval D**4** between the plurality of second units **30** adjacent to each other in the Y-axis direction.

[0068] The pair of second side shells **34** is formed of a conductive metal plate, for example, and is a coupling member that couples the plurality of blocks **330** of the second housing **33** along the Y-axis direction. The pair of second side shells **34** is disposed at both ends of the plurality of blocks **330** in the X-axis direction. The pair of second side shells **34** engages with each of the plurality of blocks **330** to couple the plurality of blocks **330**.

[0069] FIG. **10**A is a perspective view illustrating an example of the appearance of the pair of second contacts **31**. FIG. **10**B is a plan view illustrating an example of the appearance of the pair of second contacts **31**. FIG. **10**C is a side view illustrating an example of the appearance of the pair of second contacts **31**. FIG. **10**D is a bottom view illustrating an example of the appearance of the pair of second contacts **31**.

[0070] As illustrated in FIG. **10**C, the pair of second contacts **31** extends along the Z-axis direction when viewed from the arrangement direction. The pair of second contacts **31** takes on a form approximating an S shape when viewed from the arrangement direction. The pair of second contacts **31** is formed by, for example, performing bending processing or the like on a punched metal plate. Each of the pair of second contacts **31** includes a second contact portion **311**, a second connection leg portion **312**, and a second coupling portion **313**.

[0071] The second contact portion **311** has a flat plate shape extending along the Z-axis direction. The second connection leg portion **312** extends along the second board **300** and is electrically connected to the signal conductors of the second board **300**. The second coupling portion **313** extends along the Z-axis direction and couples the second contact portion **311** and the second connection leg portion **312**.

[0072] FIG. **11**A is a perspective view illustrating an example of the appearance of the second shell **32**. FIG. **11**B is a plan view illustrating an example of the appearance of the second shell **32**. FIG. **11**C is a side view illustrating an example of the appearance of the second shell **32**. FIG. **11**D is a bottom view illustrating an example of the appearance of the second shell **32**.

[0073] As illustrated in FIG. **11**B and FIG. **11**D, the second shell **32** has a so-called U-shape or H-shape when viewed from the fitting direction. The second shell **32** is formed by, for example, performing bending processing or the like on a punched metal plate. The second shell **32** includes a pair of second side surfaces **321** facing each other and a second coupling surface **322** coupling the pair of second side surfaces **321**. The second shell **32** does not have a surface facing the second coupling surface **322**.

[0074] As illustrated in FIG. **11**A, the pair of second side surfaces **321** has a flat plate shape extending in the Y-axis direction and the Z-axis direction. Each of the pair of second side surfaces **321** includes a ground connection portion **321***a* that faces the second board **300** and is formed along the Y-axis direction. The ground connection portion **321***a* is electrically connected to the ground conductors of the second board **300**. The pair of second side surfaces **321** includes a pair of protrusions **323** protruding away from the second board **300**.

[0075] The pair of protrusions **323** functions as a so-called leaf spring. The pair of protrusions **323** is curved so as to be spaced apart from each other in the X-axis direction as being away from the second board **300** along the fitting direction. Each of the pair of protrusions **323** includes a base portion **323***a*, a contact portion **323***b*, and a tip portion **323***c*. The base portion **323***a*, the contact portion **323***b*, and the tip portion **323***c* are provided in this order away from the second board **300**. The base portions **323***a* are inclined so as to be spaced apart from each other. The contact portions **323***b* are portions that are furthest apart from each other. The tip portions **323***c* are inclined so as to be closer to each other. The protrusion **323** is formed to be gradually wider from the tip portion **323***c* toward the base portion **323***a*. That is, the protrusion **323** is wider toward the first board **200**.

[0076] The second coupling surface **322** has a flat plate shape extending in the X-axis direction and the Z-axis direction. The second coupling surface **322** couples the pair of second side surfaces **321** at both ends of the second coupling surface **322** in the X-axis direction. The second coupling surface **322** includes a ground connection portion **322***a* that faces the second board **300** and is formed along the X-axis direction. The ground connection portion **322***a* is electrically connected to the ground conductors of the second board **300**.

[0077] FIG. **12**A is a perspective view illustrating an example of the appearance of the second unit **30**. FIG. **12**B is a plan view illustrating an example of the appearance of the second unit **30**. FIG. **12**C is a side view illustrating an example of the appearance of the second unit **30**. FIG. **12**D is a bottom view illustrating an example of the appearance of the second unit **30**.

[0078] The pair of second side surfaces **321** faces each other with the pair of second contacts **31** interposed therebetween in the arrangement direction of the pair of second contacts **31**. The pair of second side surfaces **321** is disposed so as to cover the pair of second contacts **31** in the arrangement direction of the pair of second contacts **31**. As illustrated in FIG. **12**B, the total length of the pair of second side surfaces **321** in the Y-axis direction is longer than the total length of the pair of second contacts **31** in the Y-axis direction. The total length of the ground connection portion **321***a* in the Y-axis direction is longer than the total length of the second connection leg portion **312** in the Y-axis direction.

[0079] The protrusions **323** face the second contact portions **311** of the pair of second contacts **31** in the arrangement direction of the pair of second contacts **31**. The protrusions **323** are disposed so as to cover the second contact portions **311** in the arrangement direction of the pair of second contacts. As illustrated in FIG. **12**C, the total length of the pair of second side surfaces **321** in the Z-axis direction is longer than the total length of the pair of second contacts **31** in the Z-axis direction.

[0080] The second housing **33** will be described with reference to FIGS. **13** and **14**. FIG. **13** is an enlarged plan view illustrating an example of a part of the appearance of the mating connector **3** in an enlarged manner. FIG. **14** is an enlarged bottom view illustrating an example of a part of the appearance of the mating connector **3** in an enlarged manner. When viewed from the fitting direction, the second housing **33** takes on a form approximating a rectangular shape. Each of the blocks **330** of the second housing **33** includes a main surface **330***a* and a back surface **330***b* that extend in a planar shape along the second board **300**.

[0081] The main surface **330***a* is a surface facing the electric connector **2** when fitted to the electric connector **2**. The back surface **330***b* is a surface facing the second board **300**. Each of the blocks **330** is formed with, for each of the plurality of second units **30**, a columnar portion **331** and an engagement groove **335** that connect the main surface **330***a* and the back surface **330***b*. The columnar portion **331** is formed with a hole **332** penetrating so as to connect the main surface **330***a* and the back surface **330***b*. The columnar portion **331** is also provided with a pair of side surfaces **333** provided so as to be orthogonal to the X-axis direction. Each of the blocks **330** includes, at both ends in the X-axis direction, fitting protrusions **334** protruding away from the second board **300** with the main surface **330***a* as a base end.

[0082] The columnar portion **331** is formed in a prismatic shape by, for example, insert molding along the arrangement of the pair of second contacts **31**. The columnar portion **331** accommodates the second coupling portions **313** of the pair of second contacts **31**, and exposes the second connection leg portions **312** of the pair of second contacts **31** from the back surface **330***b*. In the hole **332**, a part of the second shell **32** is accommodated. The hole **332** accommodates a part of the second coupling surface **322**, and exposes the ground connection portion **322***a* from the back surface **330***b*. The pair of side surfaces **333** comes into contact with the pair of second side surfaces **321** of the second shell **32**. The fitting protrusion **334** has a columnar shape. The fitting protrusion **334** has a curved surface **334***a* connecting the tip and the side surface. The length from the main surface **330***a* to the tip of the fitting protrusion **334** may be longer than the length from the main surface **330***a* to the tip (tip portion **323***c*) of the second unit **30**. The fitting protrusion **334** may protrude beyond the second unit **30**. The fitting protrusion **334** is fitted into the fitting hole **235** when fitted to the electric connector **2**. The fitting protrusion **334** and the fitting hole **235** function as guides at the time of fitting the electric connector **2** and the mating connector **3**. The curved surface **334***a* of the fitting protrusion **334** may slide when coming into contact with the main surface **230***a* in the vicinity of the fitting hole **235** of the first housing **23** to align the fitting protrusion **334** and the fitting hole **235**. In the engagement groove **335**, the second side shell **34** is accommodated. One or more engagement grooves **335** may be provided per block **330**. For example, a plurality of engagement grooves **335** is provided at predetermined intervals in the Y-axis direction.

[0083] FIG. **15** is a side view illustrating an example of coupling by the second side shell **34**. The second side shell **34** has a comb-like shape. The second side shell **34** is formed of, for example, a metal plate. The second side shell **34** includes a coupling plate **341** and a protrusion **342**. The coupling plate **341** has an elongated plate-like shape and extends in a planar shape in the Y-axis direction. The protrusions **342** protrude toward the engagement grooves **335** of the plurality of blocks **330** in the fitting direction. The protrusions **342** are provided so as to correspond to the number of engagement grooves **335**. For example, the plurality of protrusions **342** is provided at predetermined intervals in the Y-axis direction. The plurality of protrusions **342** is accommodated in the plurality of engagement grooves **236** by press-fitting for each axis K**2** along the fitting direction.

Fitted State

[0084] A state where the electric connector **2** and the mating connector **3** are fitted to each other (hereinafter, the state is referred to as a "fitted state") will be described in detail with reference to FIGS. **16** to **19**. FIG. **16** is a perspective view illustrating an example of the appearance of the first unit **20** and the second unit **30** in a fitted state. FIG. **17** is a plan view illustrating an example of the appearance of the first unit **20** and the second unit **30** in a fitted state. FIG. **18** is a cross-sectional view illustrating a cross section taken along the line xviii-xviii of FIG. **17**. FIG. **19** is a cross-sectional view illustrating a cross section taken along the line xix-xix of FIG. **17**. In FIGS. **16** to **19**, the first housing **23**, the second housing **33**, the first board **200**, and the second board **300** are not illustrated.

[0085] As illustrated in FIG. **16**, in the fitted state, the first coupling surface **222** does not face the second connection leg portion **312**. The first coupling surface **222** opens the second connection leg portion **312**. The second coupling surface **322** does not face the first connection leg portion **212**. The second coupling surface **322** opens the first connection leg portion **212**. In the fitted state, the first coupling surface **222** and the second coupling surface **322** do not face each other when viewed from the direction orthogonal to the fitting direction and the arrangement direction of the pair of first contacts **21**. That is, the first coupling surface **222** and the second coupling surface **322** do not face each other in the Y-axis direction. In the fitted state, the first coupling surface **222** is not in contact with the second shell **32**.

[0086] As illustrated in FIG. **17**, when viewed from the fitting direction in the fitted state, a region R surrounded by the first shell **22** and the second shell **32** is formed. When viewed from the fitting direction in the fitted state, the region R is formed by fitting the first coupling surface **222** of the first shell **22** and the second coupling surface **322** of the second shell **32** so as to face each other. When viewed from the fitting direction in the fitted state, the pair of first contacts **21** is electrically connected to the signal conductors of the first board **200** outside the region R surrounded by the first shell **22** and the second shell **32**. When viewed from the fitting direction in the fitted state, the pair of first contacts **21** passes between the second coupling surface **322** and the first board **200**, extends outside the region R surrounded by the first shell **22** and the second shell **32**, and is electrically connected to the signal conductors of the first board **200**. When viewed from the fitting direction in the fitted state, the pair of second contacts **31** is electrically connected to the signal conductors of the second board **300** in the region R.

[0087] When viewed from the fitting direction in the fitted state, the pair of first contacts **21** and the pair of second contacts **31** come into contact with each other in the region R. As illustrated in FIG. **18**, in the fitted state, the pair of first contacts **21** and the pair of second contacts **31** come into contact with each other to form a contact touching portion C. The contact touching portion C is a

contact point between the first contact portions **211** of the pair of first contacts **21** and the second contact portions **311** of the pair of second contacts **31**. In the fitted state, the first contact portions **211** do not face the second coupling surface **322** with the second contact portions **311** interposed therebetween.
[0088] Soldering is performed on a contact point of the first connection leg portion **212** with the first board **200**. A distance D**5** between the first board **200** and the contact touching portion C is smaller than a distance D**6** between the contact point of the first connection leg portion **212** with the first board **200** and the contact touching portion C. This may prevent so-called solder rise in which the solder applied between the first connection leg portion **212** and the first board **200** reaches the contact touching portion C.
[0089] Returning to FIG. **17**, when viewed from the fitting direction in the fitted state, the first shell **22** and the second shell **32** surround the contact touching portion C where the pair of first contacts **21** and the pair of second contacts **31** come into contact with each other around the axis L. When viewed from the fitting direction in the fitted state, the pair of first side surfaces **221**, the pair of second side surfaces **321**, the first coupling surface **222**, and the second coupling surface **322** surround the contact touching portion C around the axis L. When viewed from the fitting direction in the fitted state, the contact touching portion C is formed in the region R surrounded by the first shell **22** and the second shell **32**.
[0090] The pair of first side surfaces **221** faces each other with the first connection leg portions **212** interposed therebetween in the arrangement direction of the pair of first contacts **21**. The pair of second side surfaces **321** faces each other with the second connection leg portions **312** interposed therebetween in the arrangement direction of the pair of second contacts **31**. In the fitted state, the pair of first side surfaces **221** faces each other with the contact touching portion C interposed therebetween in the arrangement direction of the pair of first contacts **21**, and the pair of second side surfaces **321** faces each other with the contact touching portion C interposed therebetween in the arrangement direction of the pair of second contacts **31**.
[0091] In the fitted state, the first connection leg portions **212** and the second connection leg portions **312** extend in directions away from each other. For example, the first connection leg portions **212** extend in the positive direction of the Y axis, and the second connection leg portions **312** extend in the negative direction of the Y axis.
[0092] As illustrated in FIG. **19**, in the fitted state, the first shell **22** comes into contact with the second shell **32** so as to sandwich the second shell **32** to form shell touching portions S. Each of the shell touching portions S is a contact point between the pair of first side surfaces **221** of the first shell **22** and the pair of second side surfaces **321** of the second shell **32**. Each of the shell touching portions S is a contact point between the pair of first side surfaces **221** and the pair of protrusions **323** of the pair of second side surfaces **321**. The shell touching portions S are closer to the first board **200** than the contact touching portions C.
[0093] The pair of protrusions **323** elastically deforms to come into contact with the pair of first side surfaces **221**. At the time of fitting, the tip portions **323***c* of the pair of protrusions **323** come into contact with the first shell **22** and slide in accordance with the inclination, thereby functioning as a guide for alignment. The contact portions **323***b* of the pair of protrusions **323** come into contact with the pair of first side surfaces **221** in a pressing manner.
Summary
[0094] The above examples include the following configurations.
[0095] (1) A connector device **1** including an electric connector **2** mounted on a first board **200**; and a mating connector **3** mounted on a second board **300** and fitted to the electric connector **2**, in which the electric connector **2** includes: a pair of conductive first contacts **21** electrically connected to a signal conductor of the first board **200**; a conductive first shell **22** electrically connected to a ground conductor of the first board **200** and partially covering a periphery of the pair of first contacts **21** around an axis L along a fitting direction between the electric connector **2** and the

mating connector **3**; and a first housing **23** configured to hold the pair of first contacts **21** and the first shell **22** in an insulated state, the mating connector **3** includes: a pair of conductive second contacts **31** electrically connected to a signal conductor of the second board **300**; a conductive second shell **32** electrically connected to a ground conductor of the second board **300** and partially covering a periphery of the pair of second contacts **31** around the axis; and a second housing **33** configured to hold the pair of second contacts **31** and the second shell **32** in an insulated state, the pair of first contacts **21** comes into contact with the pair of second contacts **31** and the first shell **22** comes into contact with the second shell **32** in a fitted state where the electric connector **2** and the mating connector **3** are fitted to each other, and the first shell **22** and the second shell **32** surround a contact touching portion C where the pair of first contacts **21** and the pair of second contacts **31** come into contact with each other around the axis when viewed from the fitting direction in the fitted state.

[0096] In the electric connector **2**, the first shell **22** partially covers the periphery of the pair of first contacts **21** around the axis. This enables simplifying the first shell **22** as compared with a case where the first shell **22** covers the entire periphery of the pair of first contacts **21** around the axis. In the mating connector **3**, the second shell **32** partially covers the periphery of the pair of second contacts **31** around the axis. This enables simplifying the second shell **32** as compared with a case where the second shell **32** covers the entire periphery of the pair of second contacts **31** around the axis. When viewed from the fitting direction in the fitted state, the first shell **22** and the second shell **32** surround the contact touching portion C around the axis. This improves the shielding performance in the fitted state.

[0097] (2) The connector device **1** according to (1), in which the first shell **22** includes a pair of first side surfaces **221** facing each other and a first coupling surface **222** coupling the pair of first side surfaces **221**, the second shell **32** includes a pair of second side surfaces **321** facing each other and a second coupling surface **322** coupling the pair of second side surfaces **321**, and the pair of first side surfaces **221**, the pair of second side surfaces **321**, the first coupling surface **222**, and the second coupling surface **322** surround the contact touching portion C around the axis when viewed from the fitting direction in the fitted state. The first shell **22** has a so-called U-shape or H-shape with the pair of first side surfaces **221** and the first coupling surface **222**. The second shell **32** has a so-called U-shape or H-shape with the pair of second side surfaces **321** and the second coupling surface **322**. Such shapes of the first shell **22** and the second shell **32** may be readily manufactured by, for example, performing bending processing or the like on a punched metal plate. This enables simplifying the first shell **22** and the second shell **32**. Further, when viewed from the fitting direction in the fitted state, the pair of first side surfaces **221**, the pair of second side surfaces **321**, the first coupling surface **222**, and the second coupling surface **322** surround the periphery of the contact touching portion C around the axis. This improves the shielding performance in the fitted state.

[0098] (3) The connector device **1** according to (2), in which the pair of second side surfaces **321** includes a pair of protrusions **323** protruding away from the second board **300**, the pair of protrusions **323** comes into contact with the pair of first side surfaces **221** in the fitted state, and a shell touching portion S where the pair of protrusions **323** and the pair of first side surfaces **221** come into contact with each other is closer to the first board **200** than the contact touching portion C. In the fitted state, the pair of protrusions **323** is in contact with the pair of first side surfaces **221** at a position closer to the first board **200** than the contact touching portion C. The distance between the second board **300** and the first board **200** is reduced and the length of the pair of protrusions **323** is increased in the direction from the second board **300** toward the first board **200**, which increases the elastic force of the pair of protrusions **323**.

[0099] (4) The connector device **1** according to any one of (1) to (3), in which when viewed from the fitting direction in the fitted state, the pair of first contacts **21** is electrically connected to the signal conductor of the first board **200** outside a region surrounded by the first shell **22** and the second shell **32**. In the region surrounded by the first shell **22** and the second shell **32**, the pair of first contacts **21** comes into contact with the pair of second contacts **31**. The pair of first contacts **21** is electrically connected to the signal conductor of the first board **200** outside the region surrounded by the first shell **22** and the second shell **32**. That is, the pair of first contacts **21** is formed both inside and outside the region surrounded by the first shell **22** and the second shell **32**. In such a pair of first contacts **21**, the length from the contact point with the pair of second contacts **31** to the contact point with the signal conductor of the first board **200** increases, so that the elastic force of the pair of first contacts **21** may be increased.

[0100] (5) The connector device **1** according to any one of (1) to (4), in which the electric connector **2** includes a plurality of first units **20** each of which includes the pair of first contacts **21** and the first shell **22**, the plurality of first units **20** is disposed along a direction orthogonal to the fitting direction and a direction orthogonal to an arrangement direction of the pair of first contacts **21**, and is disposed along the arrangement direction of the pair of first contacts **21**, the mating connector **3** includes a plurality of second units **30** each of which includes the pair of second contacts **31** and the second shell **32**, and the plurality of second units **30** is disposed along a direction orthogonal to the fitting direction and a direction orthogonal to an arrangement direction of the pair of second contacts **31**, and is disposed along the arrangement direction of the pair of second contacts **31**. In the fitted state, the shielding performance may be achieved for each of the plurality of first units **20** and the plurality of second units **30**. As a result, the intervals D**1** and D**2** between the first units **20** and the intervals D**3** and D**4** between the second units **30** may be reduced, which enables the electric connector **2** and the mating connector **3** to achieve higher density.

[0101] (6) The connector device **1** according to (2) or (3), in which the pair of first contacts **21** includes a first connection leg portion **212** extending along the first board **200** and electrically connected to the signal conductor of the first board **200**, the pair of second contacts **31** includes a second connection leg portion **312** extending along the second board **300** and electrically connected to the signal conductor of the second board **300**, and in the fitted state, the first connection leg portion **212** and the second connection leg portion **312** extend in directions away from each other. The first connection leg portion **212** and the second connection leg portion **312** extend in directions away from each other. Since the first connection leg portion **212** and the second connection leg portion **312** are spaced apart from each other, it may prevent the influence that the first connection leg portion **212** and the second connection leg portion **312** have on each other.

[0102] (7) The connector device **1** according to (6), in which the pair of first side surfaces **221** faces each other with the first connection leg portion **212** interposed therebetween in an arrangement direction of the pair of first contacts **21**, the pair of second side surfaces **321** faces each other with the second connection leg portion **312** interposed therebetween in an arrangement direction of the pair of second contacts **31**, and in the fitted state, the pair of first side surfaces **221** faces each other with the contact touching portion C interposed therebetween in the arrangement direction of the pair of first contacts **21**, and the pair of second side surfaces **321** faces each other with the contact touching portion C interposed therebetween in the arrangement direction of the pair of second contacts **31**. This improves the shielding performance in the arrangement direction of the pair of first contacts **21** and the pair of second contacts **31**.

[0103] (8) The connector device **1** according to (6) or (7), in which, in the fitted state, the first coupling surface **222** does not face the second connection leg portion **312**, and the second coupling surface **322** does not face the first connection leg portion **212**. In the fitted state, the first coupling surface **222** opens the second connection leg portion **312**, and the second coupling surface **322** opens the first connection leg portion **212**. This improves the visibility of the first connection leg portion **212** and the second connection leg portion **312**.

[0104] (9) The connector device **1** according to (2) or (3), in which, when viewed from the fitting direction in the fitted state, the pair of first contacts **21** passes between the second coupling surface

322 and the first board **200**, extends outside a region surrounded by the first shell **22** and the second shell **32**, and is electrically connected to the signal conductor of the first board **200**. The pair of first contacts **21** is formed to extend between the second coupling surface **322** and the first board **200**. That is, the pair of first contacts **21** is formed both inside and outside the region surrounded by the first shell **22** and the second shell **32**. In such a pair of first contacts **21**, the length from the contact point with the pair of second contacts **31** to the contact point with the signal conductor of the first board **200** increases, so that the elastic force of the pair of first contacts **21** may be increased.

[0105] (10) The connector device **1** according to (2) or (3), in which, in the fitted state, the first coupling surface **222** and the second coupling surface **322** do not face each other when viewed from a direction orthogonal to the fitting direction and a direction orthogonal to an arrangement direction of the pair of first contacts **21**. In this case, the visibility of the pair of first contacts **21** and the pair of second contacts **31** is increased.

[0106] (11) The connector device **1** according to any one of (1) to (10), in which the first housing **23** is formed with a fitting hole **235** penetrating the first housing **23** so as to connect a main surface **230***a* facing the mating connector **3** when fitted to the mating connector **3** and a back surface **230***b* facing the first board **200**, the second housing **33** is formed with a fitting protrusion **334** protruding away from the second board **300** with a main surface **330***a* facing the electric connector **2** when fitted to the electric connector **2** as a base end, and the fitting hole **235** and the fitting protrusion **334** are fitted to each other in the fitted state. In this case, the fitting protrusion **334** and the fitting hole **235** function as guides at the time of fitting the electric connector **2** and the mating connector **3**. As a result, the reliability of the contact between the first contact **21** and the second contact **31** may be increased.

[0107] (12) An electric connector **2** mounted on a first board **200** and fitted to a mating connector **3** mounted on a second board **300**, the electric connector **2** including: a pair of first contacts **21** electrically connected to a signal conductor of the first board **200**; a first shell **22** electrically connected to a ground conductor of the first board **200** and partially covering a periphery of the pair of first contacts **21** around an axis L along a fitting direction between the electric connector **2** and the mating connector **3**; and a housing (first housing **23**) configured to hold the pair of first contacts **21** and the first shell **22** in an insulated state, in which the pair of first contacts **21** comes into contact with a pair of second contacts **31** of the mating connector **3** and the first shell **22** comes into contact with a second shell **32** of the mating connector **3** in a fitted state with the mating connector **3**, and the first shell **22** and the second shell **32** surround a contact touching portion C where the pair of first contacts **21** and the pair of second contacts **31** come into contact with each other around the axis when viewed from the fitting direction in the fitted state.

[0108] In the electric connector **2**, the first shell **22** partially covers the periphery of the pair of first contacts **21** around the axis. This enables simplifying the first shell **22** as compared with a case where the first shell **22** covers the entire periphery of the pair of first contacts **21** around the axis. When viewed from the fitting direction in the fitted state with the mating connector **3**, the first shell **22** and the second shell **32** surround the contact touching portion C around the axis. This improves the shielding performance in the fitted state.

MODIFIED EXAMPLES

[0109] It is to be understood that not all aspects, advantages and features described herein may necessarily be achieved by, or included in, any one particular example. Indeed, having described and illustrated various examples herein, it should be apparent that other examples may be modified in arrangement and detail.

[0110] In the above-described example, an example in which the electric connector **2** is a receptacle connector has been described, but the electric connector **2** may be a plug connector. The mating connector **3** may be a receptacle connector. Even in this example, according to the connector device **1**, the electric connector **2**, and the mating connector **3** of the present disclosure, the shell may be simplified.

[0111] In the above-described example, the plurality of blocks **230** and the plurality of blocks **330** are disposed in the same orientation, but the present disclosure is not limited thereto. For example, the plurality of blocks **230** and the plurality of blocks **330** may be disposed in different orientations. The plurality of blocks **230** and the plurality of blocks **330** may include blocks having different configurations. For example, the plurality of blocks **230** may hold the plurality of first contacts **21** without holding the first shell **22**. The plurality of blocks **330** may hold the plurality of second contacts **31** without holding the second shell **32**.
[0112] FIG. **20** is a perspective view illustrating an example of the appearance of a connector device **1**A according to a modification. The connector device **1**A is different from the connector device **1** in that the connector device **1**A includes an electric connector **2**A and a mating connector **3**A. Hereinafter, differences of the connector device **1**A from the connector device **1** will be described. The electric connector **2**A includes a first frame **24**A instead of the first side shell **24**. The mating connector **3**A includes a second frame **34**A instead of the second side shell **34**.
[0113] The first frame **24**A is a coupling member that couples the plurality of blocks **230** of the first housing **23** along the Y-axis direction. The first frame **24**A has a frame shape surrounding the plurality of blocks **230** in the X-axis direction and the Y-axis direction. The first frame **24**A engages with each of the plurality of blocks **230** to couple the plurality of blocks **230**. The first frame **24**A is made of resin, for example. For example, the first frame **24**A is formed by positioning the plurality of blocks **230** side by side, then housing the plurality of blocks **230** in a mold having cavities shaped like the first frame **24**A, and injecting resin into the cavities.
[0114] The second frame **34**A is a coupling member that couples the plurality of blocks **330** of the second housing **33** along the Y-axis direction. The second frame **34**A has a frame shape surrounding the plurality of blocks **330** in the X-axis direction and the Y-axis direction. The second frame **34**A engages with each of the plurality of blocks **330** to couple the plurality of blocks **330**. The second frame **34**A is made of resin, for example. For example, the second frame **34**A is formed by positioning the plurality of blocks **330** side by side, then housing the plurality of blocks **330** in a mold having cavities shaped like the second frame **34**A, and injecting resin into the cavities.
[0115] FIG. **21**A is a perspective view illustrating an example of the appearance of the first unit **20**A according to a modification. FIG. **21**B is a plan view illustrating an example of the appearance of the first unit **20**A according to the modification. FIG. **21**C is a side view illustrating an example of the appearance of the first unit **20**A according to the modification. FIG. **21**D is a bottom view illustrating an example of the appearance of the first unit **20**A according to the modification. The first unit **20**A is different from the first unit **20** in that the first unit **20**A includes a first shell **22**A instead of the first shell **22**. Differences of the first unit **20**A from the first unit **20** will be described below. The first unit **20**A includes a pair of first upper surfaces **223**.
[0116] The pair of first upper surfaces **223** has a flat plate shape extending toward each other with upper ends of the pair of first side surfaces **221** as base ends. For example, the pair of first upper surfaces **223** is formed so as to bend from the upper end of one first side surface **221** toward the other first side surface **221** and face each other. The pair of first upper surfaces **223** is disposed above the first coupling portions **213** of the pair of first contacts **21** in the fitting direction. The pair of first upper surfaces **223** faces the first board **200** with the first coupling portion **213** interposed therebetween. The pair of first upper surfaces **223** is not positioned above the first contact portions **211** in the fitting direction. That is, the pair of first upper surfaces **223** does not face the first contact portions **211**. According to such a configuration, the shielding property of the first unit **20**A is increased.

Claims

1. A connector device comprising: an electric connector mounted on a first board; and a mating connector mounted on a second board and fitted to the electric connector, wherein the electric connector comprises: a pair of conductive first contacts electrically connected to a signal conductor of the first board, a conductive first shell electrically connected to a ground conductor of the first board and partially covering a periphery of the pair of first contacts around an axis along a fitting direction between the electric connector and the mating connector, and a first housing configured to hold the pair of first contacts and the first shell in an insulated state, the mating connector comprises: a pair of conductive second contacts electrically connected to a signal conductor of the second board, a conductive second shell electrically connected to a ground conductor of the second board and partially covering a periphery of the pair of second contacts around the axis, and a second housing configured to hold the pair of second contacts and the second shell in an insulated state, wherein the pair of first contacts comes into contact with the pair of second contacts and the first shell comes into contact with the second shell in a fitted state where the electric connector and the mating connector are fitted to each other, and wherein the first shell and the second shell surround a contact touching portion where the pair of first contacts and the pair of second contacts come into contact with each other around the axis when viewed from the fitting direction in the fitted state.

2. The connector device according to claim 1, wherein when viewed from the fitting direction in the fitted state, the pair of first contacts is electrically connected to the signal conductor of the first board outside a region surrounded by the first shell and the second shell.

3. The connector device according to claim 1, wherein the first housing is formed with a fitting hole penetrating the first housing so as to connect a main surface of the first housing facing the mating connector when fitted to the mating connector and a back surface facing the first board, wherein the second housing is formed with a fitting protrusion that protrudes away from the second board using a main surface of the second housing facing the electric connector when fitted to the electric connector as a base end, and wherein the fitting hole and the fitting protrusion are fitted to each other in the fitted state.

4. The connector device according to claim 1, wherein the first shell covers at least three sides of the periphery of the pair of first contacts, and wherein the three sides include a first side disposed along a first direction which is an arrangement direction of the pair of first contacts, a second side disposed along the first direction and facing the first side, and a third side disposed along a second direction which is a direction orthogonal to the first direction and the fitting direction.

5. The connector device according to claim 1, wherein the first shell comprises a pair of first side surfaces facing each other and a first coupling surface coupling the pair of first side surfaces, wherein the second shell comprises a pair of second side surfaces facing each other and a second coupling surface coupling the pair of second side surfaces, and wherein the pair of first side surfaces, the pair of second side surfaces, the first coupling surface and the second coupling surface surround the contact touching portion around the axis when viewed from the fitting direction in the fitted state.

6. The connector device according to claim 5, wherein the pair of second side surfaces comprises a pair of protrusions that protrude away from the second board, wherein the pair of protrusions comes into contact with the pair of first side surfaces in the fitted state, and wherein a shell touching portion where the pair of protrusions and the pair of first side surfaces come into contact with each other is closer to the first board than the contact touching portion.

7. The connector device according to claim 5, wherein, when viewed from the fitting direction in the fitted state, the pair of first contacts passes between the second coupling surface and the first board, extends outside a region surrounded by the first shell and the second shell, and is electrically connected to the signal conductor of the first board.

8. The connector device according to claim 5, wherein, in the fitted state, the first coupling surface and the second coupling surface do not face each other when viewed from a direction orthogonal to the fitting direction and an arrangement direction of the pair of first contacts.

9. The connector device according to claim 5, wherein the pair of first contacts comprises a first connection leg portion extending along the first board and electrically connected to the signal conductor of the first board, wherein the pair of second contacts comprises a second connection leg portion extending along the second board and electrically connected to the signal conductor of the second board, and wherein in the fitted state, the first connection leg portion and the second connection leg portion extend in directions away from each other.

10. The connector device according to claim 9, wherein, in the fitted state, the first coupling surface does not face the second connection leg portion, and the second coupling surface does not face the first connection leg portion.

11. The connector device according to claim 9, wherein the pair of first side surfaces faces each other with the first connection leg portion interposed therebetween in a first direction which is an arrangement direction of the pair of first contacts, wherein the pair of second side surfaces faces each other with the second connection leg portion interposed therebetween in the first direction, and wherein in the fitted state, the pair of first side surfaces faces each other with the contact touching portion interposed therebetween in the first direction, and the pair of second side surfaces faces each other with the contact touching portion interposed therebetween in the first direction.

12. The connector device according to claim 11, wherein, in the fitting direction, a length of the pair of second side surfaces is longer than a length of the pair of second contacts.

13. The connector device according to claim 11, wherein, in a second direction which is a direction orthogonal to the first direction and the fitting direction, a length of the pair of first side surfaces is longer than a length of the pair of first contacts.

14. The connector device according to claim 13, wherein the pair of first side surfaces comprises a ground connection portion that faces the first board and is formed along the second direction, and wherein, in the second direction, a length of the ground connection portion is longer than a length of the pair of first contacts.

15. The connector device according to claim 13, wherein the first coupling surface comprises a ground connection portion that face the first board and is formed along the first direction, and wherein, in the first direction, a length of the ground connection portion is longer than a distance between the pair of first contacts.

16. The connector device according to claim 1, wherein the electric connector comprises a plurality of first units each including the pair of first contacts and the first shell, wherein the plurality of first units is disposed along a first direction which is the arrangement direction of the pair of first contacts, and is disposed along a second direction which is a direction orthogonal to the first direction and the fitting direction, wherein the mating connector comprises a plurality of second units each including the pair of second contacts and the second shell, and wherein the plurality of second units is disposed along the first direction and the second direction.

17. The connector device according to claim 16, wherein the interval between the plurality of first units adjacent to each other in the first direction is smaller than the interval between the plurality of first units adjacent to each other in the second direction.

18. The connector device according to claim 16, wherein the first housing comprises a plurality of blocks that hold the plurality of first units disposed along the first direction and are arranged side by side along the second direction.

19. The connector device according to claim 18, further comprising a conductive first side shell disposed at both ends of each block of the plurality of blocks in the first direction and connecting the plurality of blocks.

20. An electric connector mounted on a first board and fitted to a mating connector mounted on a second board, the electric connector comprising: a pair of conductive first contacts electrically connected to a signal conductor of the first board; a conductive first shell electrically connected to a ground conductor of the first board and partially covering a periphery of the pair of first contacts around an axis along a fitting direction between the electric connector and the mating connector; and a housing configured to hold the pair of first contacts and the first shell in an insulated state, wherein the pair of first contacts comes into contact with a pair of conductive second contacts of the mating connector and the first shell comes into contact with a conductive second shell of the mating connector in a fitted state with the mating connector, and wherein the first shell and the second shell surround a contact touching portion where the pair of first contacts and the pair of second contacts come into contact with each other around the axis when viewed from the fitting direction in the fitted state.